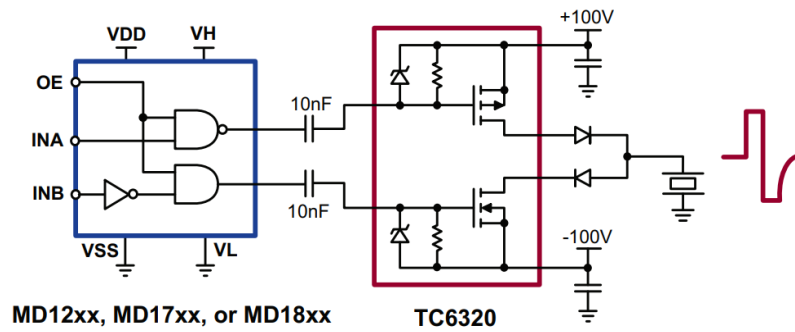
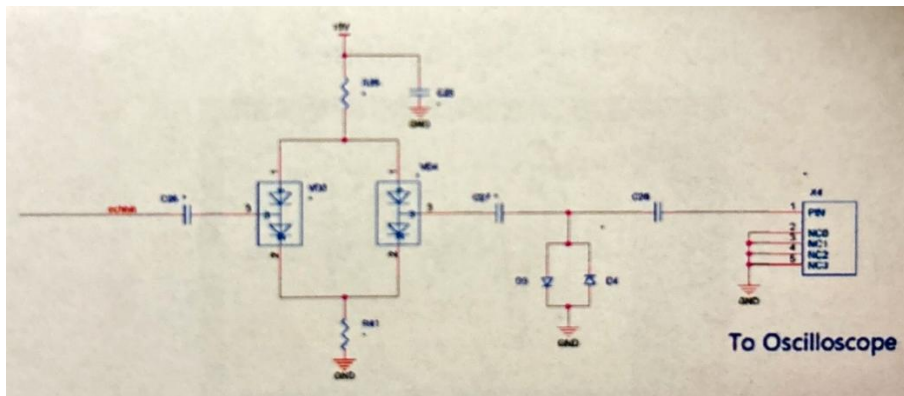
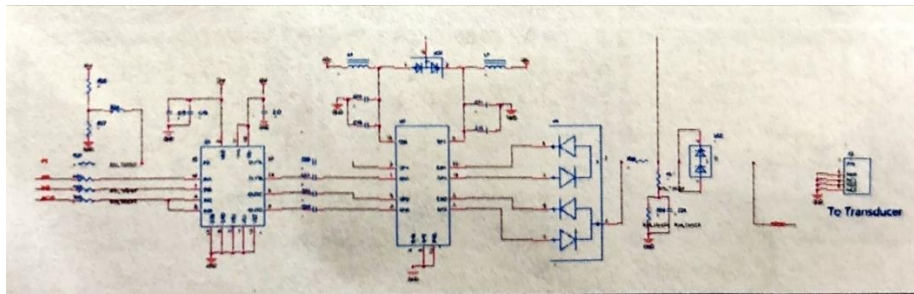


AFE Board



Pulser and driver

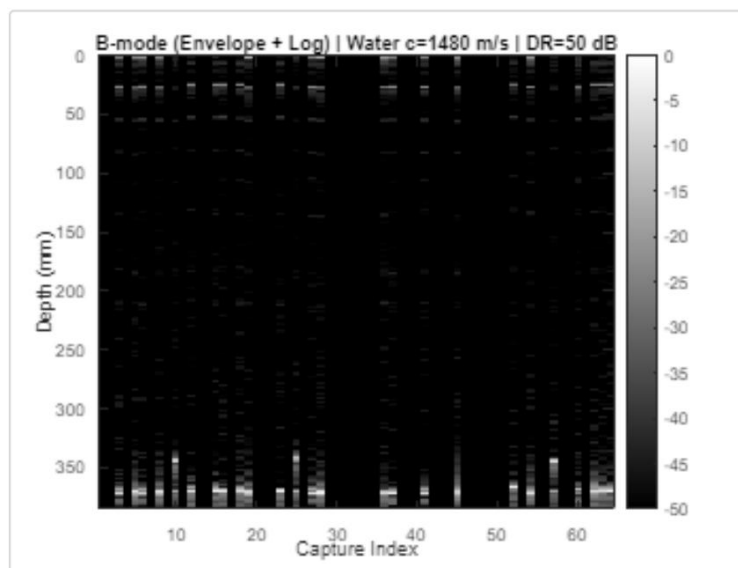
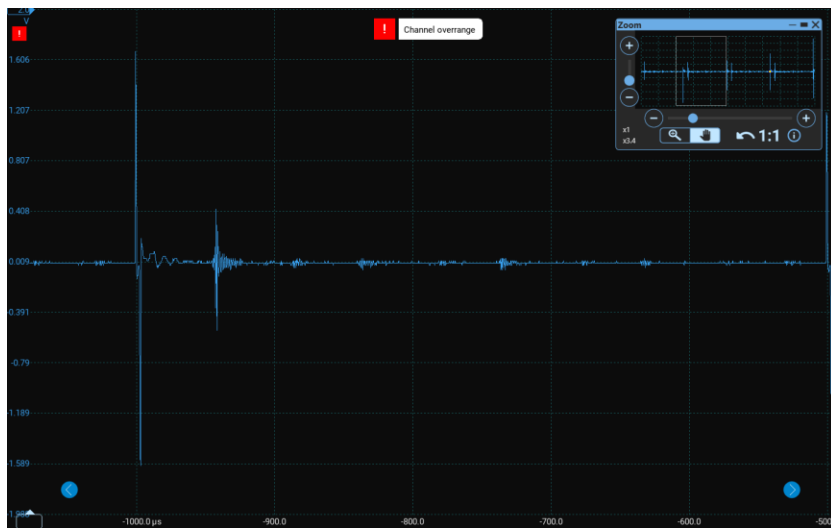
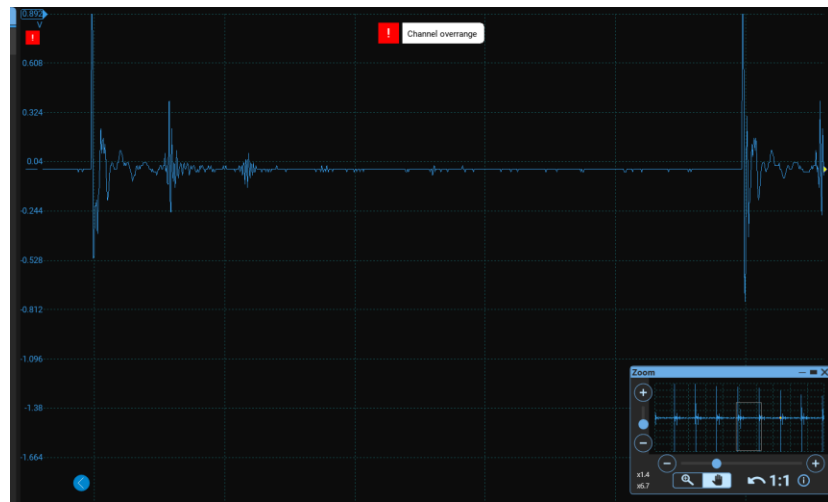


High-voltage ultrasonic transmit and receive front-end schematic. An MD12xx driver generates complementary logic signals that drive a TC6320 MOSFET stage to produce a ± 100 V bipolar excitation pulse. The receive chain includes protection diodes, AC coupling, and signal conditioning circuitry to safely route reflected echoes to the oscilloscope.



Custom pulser and analog front-end (AFE) integrating the high voltage transmit stage, transducer interface, receive protection network, amplification components, and power regulation.

Image Generation:



This image is a **B-mode ultrasound reconstruction** generated by stacking multiple A-scans and applying envelope detection and logarithmic compression.

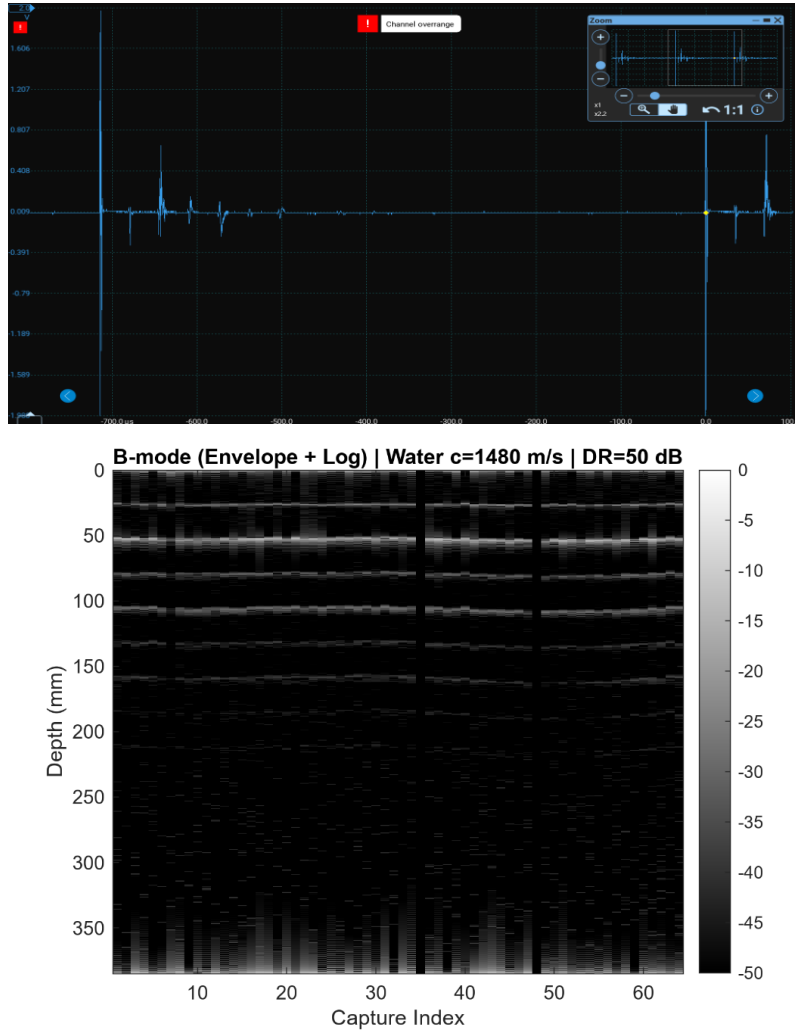
What the Axes Represent

- **X-axis (Capture Index):** Each column corresponds to one A-scan acquired at a different lateral position.
- **Y-axis (Depth in mm):** Depth calculated from time-of-flight using $c = 1480$ m/s(water).
- **Color scale (dB):** Log-compressed echo amplitude with a 50 dB dynamic range. Brighter regions indicate stronger reflections.

What the Image Shows Physically

1. **Strong shallow reflections (top ~40–80 mm):**
These horizontal bright bands correspond to strong planar reflectors in the water tank, likely the bottom surface of the pitcher and possibly internal reverberations.
2. **Uniform horizontal structure:**
The reflectors remain nearly constant across capture index, indicating a flat reflector rather than a laterally varying object.
3. **Dark vertical region (around mid capture index):**
This suggests a lower-amplitude acquisition at that lateral position, possibly due to slight misalignment, trigger variation, or gain fluctuation.
4. **Deep region (~300–380 mm):**
The bright vertical streaks near the bottom are noise amplified by log compression and depth-dependent gain compensation. This region is beyond the main reflector and primarily represents system noise.

Calibration:

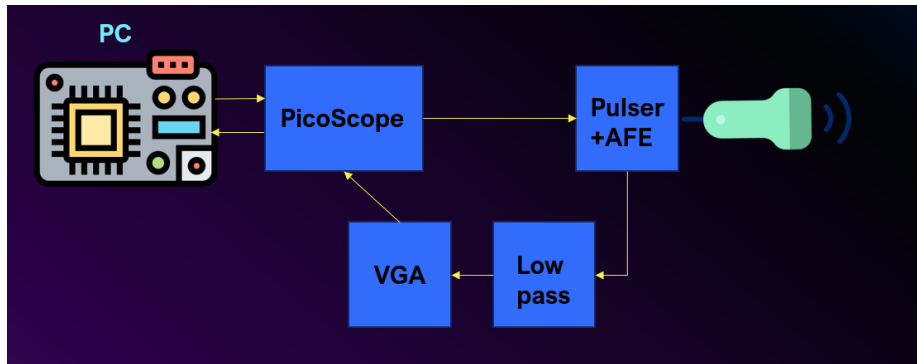


This B-mode image represents a calibration scan in water, where multiple A-scans are stacked laterally and processed using envelope detection and log compression. The prominent horizontal bright lines correspond to planar reflectors at fixed depths, most notably the bottom surface of the pitcher. Depth is calculated from time-of-flight using

$$\text{Depth} = \frac{c \cdot t}{2}$$

where $c = 1480$ m/s for water and the factor of $1/2$ accounts for round-trip propagation. The uniform horizontal bands across capture index confirm that the reflector is flat and that lateral alignment and timing are consistent. Additional parallel lines are reverberation artifacts caused by multiple reflections between the transducer and bottom surface, which are expected in calibration measurements. Overall, the clear, stable reflector bands validate proper depth conversion, envelope detection, and system calibration.

How AFE is implanted in the chain + Image processing pipeline:



PC → PicoScope → Pulser → Transducer → AFE → VGA → PicoScope → MATLAB → Image

Transmit Path

- **PC → PicoScope:** Generates trigger signal
- **Pulser (MD12xx + TC6320):** Produces ±100 V bipolar excitation pulse
- **Transducer:** Converts electrical pulse → acoustic wave

Receive Path

- **Transducer:** Echo → electrical signal
- **AFE (Protection + Low pass):** Protects circuitry and filters noise
- **VGA:** Amplifies weak echoes (depth compensation)
- **PicoScope:** Digitizes RF signal

Digital Processing (MATLAB)

- Envelope detection
- Log compression
- Depth conversion

$$\text{Depth} = \frac{ct}{2}, c = 1480 \text{ m/s}$$

- Stack A-scans → B-mode image